

Configuration-level split aliasing in object-detection benchmarks: what the reported split is selected on, and how much of the selection survives an independent split

Abstract

Object-detection papers report a number from a split they call held out. We audit that premise, in three results.

1. The convention is broken in our own corpora. 6 of 14 corpora give the reported number no independent held-out split: the checkpoint was selected there. Training and evaluation do not overlap — **checkpoint selection**, not contamination.

2. The direction varies by corpus. The premium is **positive in all four cells** (§5.1; the baseline arm gains more), the **paired Δ gap** changing sign by corpus (**not the selection term**); a declared, **descriptive** 13-cell family leaves **3** after BH; **five current-generation coders** agree at κ **0.824** and, re-verifying each row, **independently reach 19 of 30 corrected cells** (eleven unanimously), **26 of 76** stay undecidable; the printed **12–14 of 19** rest on **one rater**, **10–11** after re-verification (§6.1, S12). **The mean direction is metric-robust; one cell's significance and sign unanimity are not** (smoke2sf +0.063 pp, $p = 0.54$, 6+/5–).

3. Partial realization on a disjoint split. Separating the selection set from the reported set, the premium is realized only in part: the **claim-level interval includes zero** ([–17.3 %, 53.2 %], $P = 0.118$); on a difference scale **–1.367 pp** (CI [–1.59, –1.15]); run-level **[8.4 %, 37.4 %]** ($P = 0.002$, a bootstrap tail proportion) and leave-one-out **9.77 %–34.16 %**; the arm-cell and run-weighted point estimates are **descriptive**, given only with intervals (Table 4).

Replication lowers it 9 %–29 % (§8); no power analysis was registered, so the nine $n = 3$ cells stay descriptive.

Keywords: evaluation validity; dataset splits; checkpoint selection; object detection; benchmark audit; reproducibility

1. Introduction

A detection paper reports mAP50-95 on a split it calls held out, implying a number that estimates performance on unseen data. The inference needs a premise the paper rarely states: the checkpoint was not chosen on that same split. Binding `val:` and `test:` to one directory breaks it silently — the pipeline

evaluates there, keeps the best-scoring checkpoint and reports its score: a **maximum**, not a sample. Surveys of the detection literature [1] trace how that number became the field's unit of comparison; this paper audits the reporting convention producing it.

That premise is audited in three places, in increasing distance from our own work: **our own corpora**, where we count how many of our corpora alias the two keys (19 YAML files, counting by file); **19 public benchmarks**, audited row by row for which split the reported number comes from; and **20 published third-party training logs**, where we measure the premium directly, in data we did not produce.

Three further things we have not seen done here. We **measure the premium's fate** under a protocol separating the selection set from the reported set (**best** and **last** checkpoints and both splits exist, so the full 2×2 grid is computable): how much of the **val**-selected premium is realized on an independent split — the measurement this paper contributes and §3.1 formalises. The protocol is the companion's and is **cited, not restated**. And we **pre-register** a generalization hypothesis, then report that it misses its criterion — a registration **also reported in the companion**.

1.1 Why the unit matters

Ask "how many of these benchmarks have a split problem?" and, unless a unit is declared, every answer is defensible and they disagree by a factor of six: 2/19 (**structural**: one path bound to both keys), 10/19 (**'release' unit**: no independent held-out test), 13/19 (**'protocol' unit**: local independent testing gated), **12–14 of the 19** (**'reported' unit** — **12** the single rater's printed coding, **14** that rater's rows after a **non-blind recoding** moved two; **five blind coders agree at κ 0.824** and, once the record is re-verified first-hand, independently reach **19 of the 30** corrected cells, eleven unanimously (**S12**); the reported number's split is non-independent). The readings above are all defensible, and §6.1's four counting units (**release**, **protocol**, **yolo_dist**, **reported**) are the **columns** they come from — including **yolo_dist**, omitted above, at **4/19**. Declaring the unit is therefore a methodological premise, not a caveat (§3, §6).

1.2 What we do not claim

We do not claim that aliasing is data contamination: the image-level intersection between the training and held-out evaluation images is **zero** (SHWD excepted,

where `train` and `val` are one directory). We do not claim a single bias direction: our paired analysis finds **two corpora where `val` understates and two where it overstates**, and **nine cells are not testable at $n = 3$** (§5.1, §7.1). Nor do we claim the convention is framework-general (§6.5) or that the audited benchmarks are wrong: they report what their configuration computes.

Positioning (one sentence). We claim neither to be first to audit split problems in detection benchmarks nor to audit image-level leakage: both are taken. We occupy a different cell — **configuration-level aliasing, a direction that varies by corpus, and the requirement that a count declare its unit**. Every number points to a local artifact; every claim narrowed after external checking is recorded as narrowed (§5.4).

Scope. All quantitative claims about selection behaviour come from **YOLO-lineage pipelines**, the only family among the six frameworks we checked (five pinned repositories plus the Ultralytics release) selecting the checkpoint by the validation metric by default (§6.5) — stated at the claim, not only in a limitations section. **The registered multi-target replication is reported in the companion paper, and its first report, criterion and verdict stay there** (§9): this paper cites it for readability and **claims none of its results**.

2. Contributions and related work

Where this paper sits. Reporting practice has itself been audited: leakage and near-duplicate contamination [3], [4], [11]; split naming and the choice of standard splits [8], [9]; experiment- reporting standards [7]; checkpoint-selection criteria [6]; and budget-dependent rankings [2]. Each audits **data, reporting, or selection; none audits a release’s configuration binding** — the object of §3–§6.

2.1 The closest work, and where this paper sits (detailed)

- > The detailed comparison — the nine prior lines of work, what each measures, and the axis on which
- > this paper differs — is **Table S45 in the Supplementary material**; it is not printed here because
- > the journal counts text inside the 20–35 page limit and the comparison is background, not a claim.
- > **§2.2 (Contributions) and §2.3 (what we do not claim) are unaffected.**

2.2 Contributions

What this paper reports first (the boundary to the companion is in §2.3): the configuration-level audit of the `val/test` convention and its selection premium (§4–5); and the **measured** realization of that premium on a disjoint split with the reporting conventions it depends on (§8.6–8.7, §11, §3.1).

1. **We quantify the magnitude and direction of configuration-level aliasing.** The selection premium (best – last) is **positive in all four cells, the baseline arm gaining more (near-definitional; §5.1; four paired 95 % CIs excluding zero, §5.1 Tier 1) — consistent with the selection-bias literature [5], not new in kind. All four cells meet the stated-n criterion (n = 10–11) and all four survive a Benjamini–Hochberg step over these four at q = 0.05 — the fourth, dota15, entered at n = 3 and was escalated to n = 10 under a criterion fixed in advance (Table S28's note) — so the convention systematically shifts arm-to-arm comparisons; its direction varies by corpus: three-way paired differencing gives 2 cells where `val` understates, 2 where it overstates and 9 not testable at n = 3 — the no-systematic-direction [10] reports, measured here on detection corpora.**

We do not claim a general understatement (§5). It appears in third-party logs too (a searched 20-log sample; §5.5) — corroboration that the component exists elsewhere, **not independent evidence** and not a field-level rate.

Note: image-level leakage magnitude is quantified by the two 2026 studies; configuration-level aliasing magnitude has not been measured before.**

2. **We measure how far the selection premium transfers.** The measurement is the contribution; the protocol that makes it readable is the companion's and is **cited, not restated** here (§8.1). On the five arm-cells measured here the `val`-selected premium is realized on the disjoint split at

23.3 % with all five arm-cells at n = 10, and it is **not a constant**: the rate falls with the validation noise of the criterion used to select (§8.6–§8.7). The identity tying it to the winner's-curse share we **adopt** [13], not claim; the closest concurrent work [12, 15] reaches the same **direction** with a different quantity, and neither reports a **ratio of difference premiums** — $(T_{\text{best}} - T_{\text{final}})/(V_{\text{max}} - V_{\text{final}})$, on disjoint splits, in detection, on our own arm-cells, with measured `val` noise as moderator. A same-estimand ratio does exist elsewhere — in genomics, where it is written as a relative bias and

measured at 10–60 % (**Supplementary S12** carries the full citation). The replicate axis is **data order** (§8.4) — the **seed-budget discipline reproduced inside this setting** rather than a new one.

3. **We disclose our own defects, and the corrections that followed** rather than removing them: our analysis' first version had non-reproducible items — pod-dependent scripts, a citation to a file not holding the quoted numbers, no written endpoint convention. §10, not deleted.

2.3 What we explicitly do not claim (shared tooling)

The companion and this paper share project-internal process protocols — a multi-model review checklist and a local/cloud scripting discipline — and neither claims originality for them: **Supplementary S6, Shared process protocols**.

3. Units: the methodological premise

"Have a split problem" is a property of a (**benchmark, counting unit**) pair, not of a benchmark. [14] **varies the definition on one file; we vary who controls the split and measure a different quantity: the share of `val`-selected premium that survives a disjoint split**. We use four units, defined by **who controls the split**:

- **`release`** — the split the benchmark's **own published release** ships: the `train:` / `val:` / `test:` entries and paths in the distributed archive. *Example*: an archive whose `data.yaml` binds `val:` and `test:` to the **same directory** records "yes" here (§4).

- **`protocol`** — the split the benchmark's **official protocol** tells you to use, which need not be the released one: a protocol may carve its own validation set out of the released training split.

Example: an aliasing release whose documented protocol still requires a held-out test split records "no" here and "yes" above — the units answer different questions.

- **`yolo\_dist`** — the split a **generic YOLO distribution package's** own configuration resolves to, i.e. what a practitioner taking the pipeline off the shelf gets. *Example*: a clean release whose community YAML points `val:` and `test:` at one directory records "yes" **only** here.

- **`reported`** — the split the number **the literature prints** came from (not the official leaderboard's own score), read from the release documentation and the

dataset's own paper. *Example*: two papers on one benchmark can differ here while agreeing under all three units above — hence the headline count uses this unit (§6.1).

A count is meaningless without its unit: the same nineteen benchmarks answer differently under each (Fig. 1).

The proposition is not ours to claim. That *the same data yield different defensible audit counts under different operational definitions* is stated independently for molecular benchmark auditing, in a 2026 preprint whose abstract closes "different defensible definitions return different answers from the same file" [14]. **The observation is the cited literature's; its instantiation in detection evaluation, and the four named units, are ours** — the units named by *who controls the split*, the 19-row evidence table, and the premium on **five arm-cells**. It is orthogonal to [15], which separates leakage by **type**; we separate the **counting subject**, so one type can still be counted under four units (§6.1).

> **Table S2** → Supplementary §S8.

The 'reported' unit is read from the release documentation and from the dataset's own paper: we did not sample papers per benchmark, so no prevalence claim is made (§11). For each of 19 benchmarks we record all four units, an evidence class and a confidence level, and recompute every percentage from the row markers under a per-unit subtotal check **and one stated 'n_a' rule (Supplementary S12)** — the full 19×4 table and the cross-check against the original audit are **Supplementary S1**, the derived counts §6.1 and its table.

The rule itself is not new. The same toolchain's own course states it normatively — "*Test set is touched once, at the end. Not for tuning.*" and "*The splits must be independent on the dimension that matters at deployment.*" [16] — so what this paper adds is not the rule but the **audit and the count** under it.

The premise we rely on throughout: a "yes" under any unit means *the reported number is not a sample from an independent split*. **What that does and does not license is stated where the counts are printed** (§6.1 and the note under its table).

3.1 Formalisation: what the four quantities are, and what they can and cannot identify

Setup. Training is a randomised map from (seed ω , recipe θ , checkpoint e) to weights, and evaluation a fixed functional on a labelled split; write $V(e)$ and $T(e)$

for a checkpoint's `val` and `test` readings and $\hat{e}$ for the `val` argmax. Sections 5–9 measure four quantities: the selection premium, Δ_{gap} , κ , the realization rate (**notation and formal statements: Supplementary S11**).

Definition — the raw gap is not identifiable from the pair alone. $G = V(\hat{e}) - T(\hat{e}) = \delta + o$ mixes a split-difficulty term with a selection-optimism term; the decomposition, **and the reading that a single-model endpoint pair is unable to separate its two terms, are [17]'s**, while its **strong form** — that *any* δ' fits the observation equally well — is **stated here as a formalisation rather than attributed to that literature. The pairing assumption is what makes Δ_{gap} meaningful**, failing by construction under aliasing so that §9 cannot be read as a clean-protocol result; the identity holds value by value on all 13 scanned cells and both escalated cells at $n = 10$ (§5.6).

Adopted identity, not a claim of this paper. With $V(e) = v(e) + \varepsilon_e$ and $T(e) = v(e) + \tau + \eta_e$, $E[\text{premium_test}] = E[\text{premium_val}] - \text{WC}$, $\text{WC} := E[\varepsilon_{\hat{e}} - \varepsilon_{e_f}] \geq 0$ — the argmax is over a **candidate set fixed in advance** and e_f is a **pre-specified member of it** (what the budget-fixed `last` gives) — so the realization rate is $1 - \text{WC} / E[\text{premium_val}] \leq 1$, with equality only if selection is driven by signal alone [13] (**Supplementary S11** states which part of that the archive can and cannot settle) — the optimizer's curse in this paper's notation, the frame for §8.6–§8.7. **Two consequences are testable on the archive** — the per-run inequality, and monotonicity in the `val` noise (**Supplementary S11**). A third form, that the premium scales as $\sigma\sqrt{2 \ln E}$, was **tested and withdrawn** (**Supplementary S11**); contemporaneous work measures $\approx 5\times$ on the same $\sigma\sqrt{2 \ln K}$ bound [18].

Corollary (the premium is mostly shape, and the peak is the worst place to stand). $\text{premium_r} = \text{shape}(E) + v_r$ holds **exactly**, $\text{shape}(E)$ being the mean curve's peak-to-final gap, so shape is a functional of the mean curve alone — **not** an independent check on σ — and $E[v_r] \geq 0$. On our 40-run four-budget series **the monotone criterion is met on the clean re-run** while the $\sigma\sqrt{2 \ln E}$ form is **outside** its pre-declared band, so that scaling **stays withdrawn**; the pre-declared shape control passes on the mean it specifies but fails per arm, and we report both readings — the with-early-stopping series carrying the pre-declared verdict — so **we do not claim an 'E' effect**. The checkpoint rule is a **reporting rule** (the top-5 mean beats the argmax on the

same 40 runs), with **three limits**: an empirical capture rate, a `val`-side cost the `argmax` does not carry, and a two-split, five-epoch-grid scope.

What is ours, and what is not. Of the statements that follow, the raw gap's definition [17], the impossibility result for endpoint-pair estimators [19] and the family-dependence procedure [20] are **machinery rather than results of ours**, and the identity is **adopted** from the selection-bias literature; **the corollary alone carries measurements of this paper**. Full statements and proofs are in **Supplementary S11**, and the detail displaced from here in **Supplementary S12**.

3.2 Six experiments, five judged criteria, on GPU machines, each with its criterion written down first

The propositions of §S11 are testable, so we ran them on rented GPUs with the **criterion fixed before the runs**; **Supplementary Table S37** gives, for each of the six experiments, the criterion it was judged against. Counts below are over **experiments**, not criteria.

The two negative outcomes and the redone design are stated as such in Supplementary Table S37 (G1 is void; **two of the five judged experiments are negative** — G1' and G2; the values behind every verdict are in **S31–S33**, the G1 family in **S12**; the reading below is stated more weakly in **Supplementary S11, section 13.3**).

We then ran the clean test of G3's epoch-budget series on **one machine with early stopping disabled**, all four budgets present: C1 is **met**, C2 falls **outside** its band, and C3 **passes** on the mean $\hat{e}/E$ it specifies but not per arm — see §3.1 and **Table S34b**; **the pre-declared verdict stays with the with-early-stopping series**. The judged criteria carry one message, **at the level of sign, not of functional form**: the selection bias moves with the amount of selection noise and with the number of selection opportunities, but not with whether the two splits are the same file — **the form itself was withdrawn** (§3.1). Two consequences do not follow: §4's configuration-level fact stands, while the stronger reading — that aliasing *adds* a measurable amount on top — does not. **The premium this paper measures is the premium of selecting on the reported split; aliasing is the configuration that guarantees it, not an additional term.**

On the shared 5-epoch grid the `val` and `test` peaks agree in only **17.5 %** of runs, and conditioning on that single observable splits the 40 runs: where they

agree the **median Δ gap is lower** (mean **+0.13 pp**, median **-0.20 pp**, $n = 7$; **post-hoc**), where they differ it is **systematically not** (mean **+0.91 pp**, $n = 33$; permutation $p = 0.009$, **post-hoc**; the median reading is $p = 0.0011$). **This is a partial predictor, not an explanation of the sign**; its limits — including the four of seven agreement runs whose gap is non-positive — and the per-run table and test are in **Supplementary S11**.

4. Finding 1: our own corpora

4.1 How common the aliasing is

In **our own packaged configurations** — the YAMLs we train from, a different object from a benchmark's own release — **6 of 14** give the reported number no independent held-out split: **AI-TOD, DOTA v1, DOTA v1.5, Mendeley face-mask, SFCHD** (five binding `val:` to `test:`) and **SHWD** (which binds `train` to `val`; **S1** row 13). Of the remaining 8, **4** keep the keys genuinely distinct and **4** have no `test:` key at all — the three groups account for all 14, and this is a **count over the configurations we hold, not a prevalence, nor the benchmarks' own releases** (audited separately, §6.1). Counting by file, **19 YAML files** alias the two keys, in five locations; on the archived manifest subset the ratio is **8 of 28**. **These three bases answer three different questions** — configurations (6 of 14), the files inside them (19), the archived manifest's rows (8 of 28) — **not three estimates of one rate**.

For the corpora whose `val` and `test` are the same directory this is not contamination: their image-level intersection with training is **exactly zero**, so the aliasing changes not *what* was evaluated but *which checkpoint* was reported, turning "selection contamination" from an ethical question into a **measurement** question. **SHWD is the exception and is not folded into that statement**: with `train` and `val` on one directory, its training and evaluation images are the same files. **It still counts among the six, for the reason the list is about: its `val` is not an independent held-out split either**.

4.2 The failure is produced by a tool, not by carelessness

A PASCAL VOC [21]→YOLO conversion script wrote two keys from one `images_dir`, so `train == val` appears **because the generator could not produce distinct values** (§6.1 records the same for Mendeley face-mask [22]). §10's first recommendation: make "are `val` and `test` the same source?" a **pipeline-level hard check** — human YAML review will not catch it.

5. Finding 2: the selection premium — magnitude, direction, and mechanism

Evidence. `eval_validity/selection_premium_local_20260916.txt`, reading **569 archived runs** from the local archive (the local version is the 2026-09-16 repair).

Definition. $\text{premium} = \text{best-epoch} - \text{final-epoch}$ mAP50-95 **within a single run**. When `val == test`, "best" was selected on the reported split, while `last` and `last5` are endpoints **no selection rule can reach**. This is the checkpoint-selection term of the run-to-run variation studied in [23] (early stopping, data order) and [24] (benchmark variance across seeds). Being a maximum over a noisy curve, it also carries the upward bias of any selected extreme [13]: what those works average over as uncertainty, we measure as a reproducible component. **The two arms.** Every cell is a paired comparison in which the **baseline** arm follows the published convention and the **strategy** arm is the intervention under test — in the cells reported here, raising the peak learning rate from **0.001 to 0.005** with everything else held fixed (batch 32, `imgsz 640`, SGD; YOLOv12n with the shape-aware IoU). The two arms run the same code and the same data.

> **Table S3** → Supplementary §S8.

5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus

The direction claim **must be reported in two tiers**: "the convention understates the effect" is wrong, and external checking refuted it (§5.4, II). **Endpoint convention (frozen):** `best - last`

within a single run, both endpoints reported for every cell; §7.2–§7.3 quantify the other convention.

Tier 1: the selection premium is positive in all four cells and the baseline arm gains more, consistent with the selection-bias literature, not new in kind; **on the monitored split that positivity is near-definitional** (best is that curve's own argmax), and **the finding we report is the paired arm difference and its test side (§8.6)**. Pairing brings **no** variance benefit: the paired *t* of 5.74 and the unpaired Welch *t* [25] of 5.68 agree — at $p = 0$, **standard arithmetic rather than an empirical finding of ours** [24]. **Permutation checks** agree with the *t*-tests at each cell's attainable floor (**Supplementary S12**).

Meta-analyses of published results have long quantified this optimism: development-set early stopping is measured to *overestimate realistic model performance*, by as much as **18.0 %** accuracy [26]; and a large Kaggle meta-analysis finds *little evidence of substantial overfitting* [27] — complementary rather than contrary, since its mechanism is holdout **reuse** and ours needs none.

Metric sensitivity. On `metrics/mAP50(B)` the mean direction holds in all four cells, but `smoke2sf` loses both its significance (**+0.063 pp**, $p = 0.54$) and its sign unanimity (6+/5−, not 11+/0−): **the mean direction is metric-robust, one cell's significance and sign unanimity are not** (full mAP50 columns and both mAP75 columns — the mAP75 val side is a **5-epoch grid**, so a rate computed from it is **upward-biased** — **Supplementary S12**).

Table 1. Tier 1 — each arm's **peak–final gap** and the **paired difference** between arms, by cell; the gap is a within-run maximum (§5.1), so the finding is the paired difference. `n` counts paired units (ten file-order replicates, one unseeded — **Supplementary S12**); both differences hold at $n = 10$. **A different run set from the transfer measurement in §8.6:** these are the 569-run local archive under the published (aliased) protocol, while that measurement's premiums come from the clean three-way runs — the two tables' premiums are different quantities, not two readings of one. Source: `verify_two_critiques_20260916.txt`; the `dota15` row's $n = 10$ escalation is in Table S28.

Cell	n (paired unit s)	peak– final gap, base	peak– final gap, strat	**paired diff**	sd	t	p	per m. *p* (floor 2/2 ⁿ)	95 % CI
---	---	---	---	---	---	---	---	---	---
smoke2sf (strong)	11	+0.79 9	+0.38 2	**+0.4 16**	0.2 40	5. 74	1.9× 10 ^{−4}	0.00 098	[+0. 255, +0.5 78]
shwd2sf (corroborating)	11	+1.03 8	+0.51 2	**+0.5 26**	0.2 07	8. 41	7.6× 10 ^{−6}	0.00 098	[+0. 386, +0.6 65]
a2d15 (directional probe)	10	+0.53 4	+0.26 2	**+0.2 72**	0.1 83	4. 69	1.1× 10 ^{−3}	0.00 391	[+0. 141, +0.4 03]
dota15	10	+0.94	+0.49	**+0.4	0.1	7.	4.9×	0.00	[+0.

(within-domain) 1 2 49** 96 23 10⁻⁵ 195 309, +0.589]

Table 2. Tier 2 — the paired difference Δ_{gap} by corpus. $\Delta_{\text{gap}} > 0$ means `val` understates the strategy arm’s effect. **Δ_{gap} is the split-gap arm difference**, not the selection effect: it decomposes into a difficulty gap and a selection term, and **Δ_{gap} ’s sign is not the selection term’s sign (Fig. S1)**. Equal split difficulty across arms is an **assumption we state**; the selection contribution is the **selection-action contrast** $\Delta_{\text{prem_val}} - \Delta_{\text{prem_test}}$. ‘**Reading**’ names **Δ_{gap} ’s sign, not the selection term’s** (smoke2sf: Δ_{gap} +0.461, selection −0.580). Sources: `xeval_analysis_20260916.txt` (the two `val`-understating cells) and `x1_tier* / teval.csv` (the two `val`-overstating cells).

Corpus	n	Δ_{gap} (pp)	p	95 % CI	Reading (** Δ_{gap} ’s sign**)
---	---	---	---	---	---
shwd2sf	10	**+0.779**	0.0053	[+0.297, +1.261]	`val` **understates**
smoke2sf	10	**+0.461**	0.0263	[+0.068, +0.855]	`val` **understates**
p_aitovis	10	**−1.226**	1.3×10 ^{−8}	[−1.370, −1.081]	`val` **overstates**
p_vistod15	10	**−3.315**	6.5×10 ^{−9}	[−3.676, −2.953]	`val` **overstates**

> **The split of Δ_{gap} is recorded for two of the four cells:** difficulty +0.759 / selection +0.020 (shwd2sf), and difficulty +1.041 / selection −0.580 (smoke2sf) — in the second the selection term is **opposite in sign to Δ_{gap}** , so the sign does not carry it, and **both intervals include zero (Supplementary S9, Fig. S1)**. The two `p_` cells’ `prem_test` was **not recorded at that time**: their per-epoch test curves were evaluated on 2026-09-30, so nothing is imputed and the values are now recorded (Supplementary S12, Table S38).

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> **The four rows below are an escalated base, not a sample:** the two `p_` cells reached `n` = 10 because their `n` = 3 Δ_{gap} had been flagged, so the 2/2 balance is a property of the escalation rule; the other nine cells are **descriptive (Supplementary S12)**. This cell is also the one where **3 runs** changed training path silently under GPU memory pressure, and why we do not call it clean is in §10, defect 10.

(Third §5 table, S5 → Supplementary §S8.)

Tier 2: on the same three-way corpora the paired difference Δ_{gap} varies in sign. The raw $M_{\text{val}}(\hat{\epsilon}) - M_{\text{test}}(\hat{\epsilon})$ is **not identifiable**; the paired difference removes the difficulty term **under the equal-difficulty assumption** (Table 2) —

$\Delta_{\text{gap}} > 0 \Rightarrow$ ‘val’ understates. Its effective base is four cells, not thirteen, and an **escalated base** rather than a random sample, for the reason Table 2’s note gives; “2 understate / 2 overstate” is therefore a statement about these **four cells, not a rate for the family.** No prospective power analysis was registered; the scan is in **Supplementary S12.**

Both escalations met the criteria fixed in advance. dota15 +0.449 pp (10+/0–); p_aitovis –1.226 pp and p_vistod15 –3.315 pp (0+/10–), permutation p = **0.001953**, the attainable floor at n = 10. The best.pt endpoint was identified, not assumed: on test it reproduces the archived sio_b_results.csv to **0.0046 pp.**

The formal claim: the convention systematically shifts **arm-to-arm comparisons** (Tier 1: four cells, CIs excluding zero); **the direction of that shift varies by corpus** (Tier 2: 2 understate / 2 overstate on the four-cell base) — **neither “a general understatement” nor “no systematic direction”.** BH at q = 0.05 **now leaves three cells significant** — p_vistod15, p_aitovis, shwd2sf [28], reversing what we reported before the escalation. **The two families are not the same test:** the **13-cell scan is descriptive**, the **confirmatory family is the four escalated cells** (declared before their runs, §3.2) — the direction statement rests on those four.

Scope (cross-framework boundary). The convention this paper audits belongs to the **YOLO lineage** — Ultralytics YOLO [16] and YOLOX [29]: at pinned commits, Detectron2 (TEST.EVAL_PERIOD = 0) [30], MMDetection [31], DETR [32] and PaddleDetection [33] **report the final epoch instead** (§6.5) — that pipeline family’s convention, **not the field’s.**

5.2 Two versions must be disclosed side by side

Two archives, two versions of this table. The main text uses the **local archive** (569 runs, n = 11); the pod root (347 runs, n = 10) is a **Supplementary S12** cross-check agreeing in sign and magnitude, not in value.

5.3 The dota15 cell: now reproducible, and one wording defect exposed

This cell was not reproducible from the release as first written, and the repair exposed a wording defect (both disclosed, §10 row 2). The repair, the value-by-value agreement and the defect are recorded in **Supplementary S12**.

5.4 Two external critiques, checked (one refuted, one judged non-identifiable)

Supplementary S12 carries the two external critiques in full. We used to write that the gap was simply unidentifiable; the external check **corrects this to** the paired-difference form, which cancels the difficulty term under the pairing assumption stated in §S11. The argument and the recomputation are in **Supplementary S12**.

5.5 The same component in third-party public logs

The same quantity appears in other people's logs — a convenience sample with no sampling frame, not a field-level rate: 20 logs from 6 sources found by repository search, median **+1.04 pp**, max **+5.65 pp**, 18/20 positive. The sign is not the informative part: the premium is a within-run maximum minus that run's own final epoch, so it is **non-negative by construction** (both non-positive logs sit at exactly **+0.00**), and the sample's content is its **magnitude**. The queries, the rest of the frame and the reason public logs **cannot establish aliasing** are in **Supplementary S12**.

Five limitations: ① the two cells escalated in §5.1 reach $n = 10$ with **local** runs and test-side evaluations, and the remaining 9 cells' **test** side was closed on 2026-09-30 on one machine; ② **prem_test** is **best – last**, while the per-epoch test curve now exists for **all thirteen** corpora (**Supplementary S12**, Table S38); ③ **last.pt**'s val recomputation differs from the final **results.csv** row by ~ 0.02 pp; ④ this section does **not** touch §6's 19-benchmark census; ⑤ the aggregate is a **ratio of small means** — **prem_val** spans **+0.605 to +1.469 pp** across the five arm-cells (Table S20) — so a change of a few hundredths of a point in the numerator moves the rate by tens of points.

5.6 Why Δgap 's sign varies by corpus: three of five candidate mechanisms are refuted by the data

Three of five candidate mechanisms are refuted, and the mechanism itself is not identifiable — the sign of Δgap is a sum of same-order terms, so no single mechanism can be read off it (§S11, Remark 3) — **but the paired**

quantity is, and is what we report. The examination, the per-cell components and the exact identity are in **Supplementary S12** and Table S8–S10.

6. Finding 3: how general this is across public benchmarks

Evidence: `benchmark_split_audit.md`, a line-by-line audit of 19 benchmark rows, plus `split_units_19rows_20260916.md` (recomputation script; no hard-coded percentages).

The nineteen audited benchmarks, each read from its own release or dataset paper: COCO [34], PASCAL VOC [21], Objects365 [35], Open Images v7 [36], DOTA v1.0 and v2.0 [37], VisDrone-DET [38], AI-TOD [39], UAVDT [40], xView [41], DIOR [42], NWPU VHR-10 [43], SHWD [44], SFCHD [45], MAFA [46], Mendeley face-mask [22], WIDER FACE [47], CrowdHuman [48] and D-Fire [49] (**per-unit markings:** Supplementary S1) — **the configurations this archive could read, not a sample of the field.** The question is **evaluation validity:** dataset-level bias auditing [50] and classifier-accuracy estimation [51] ask whether a reported number measures what it is taken to measure, but neither audits a release's configuration binding.

6.1 The result under four units (single-rater judgements)

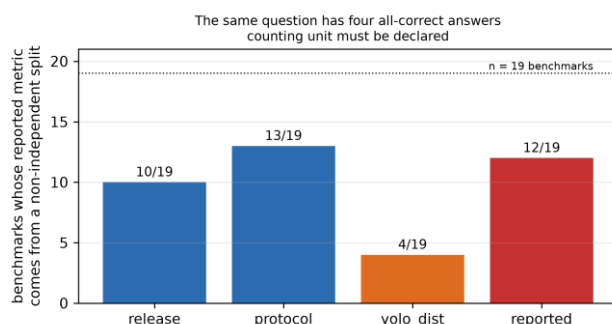


Fig. 1. The four counting units on the same nineteen benchmarks — the **printed single-rater** counts (10/19 release, 13/19 protocol, 4/19 yolo_dist, 12/19 reported); the re-verified tallies are Table 3's second column.

Table 3. The four counting units, with denominators named (§3). **Co-headline (2026-10-10):** the same four units were re-coded blind by **five current-generation coders** (κ 0.824; **Supplementary S12**), and re-verifying every row first-hand changed **30 cells** — so the table prints **both tallies**: the frozen single-rater counts, and beside them the same four units **re-tallied from the marking table after the 2026-10-09 first-party correction record**, under the **same per-column rule** (the value-to-count mapping is in **Supplementary S12**). Source:

split_units_19rows_20260916.md. Rows marked `n\_a` enter the denominator, not the numerator; alternative readings of the same rows are 12/19 for release, 12/15 the 19 less the four undecidable (n_a is two rows), and 13/19 for reported under the COCO-val reading. The `clean` row is a reading: the audit counted 5/19, a later pass moved two, so this table reads 3/19. The `reported` row's two denominators: 12-14 of the 19, and 12/15 = 80 % over the fifteen neither n_a nor undecidable (4 rows are evidence-insufficient). Both ranges span two readings, not intervals, on a single rater's judgement. Independent re-coding reproduces it only at slight-to-fair agreement (Supplementary S12). A model-assisted, non-blind second coding is reliability evidence, not a replacement (Supplementary S12).

Unit	non-independent: printed (single rater)		non-independent: after first-party re-verification	of decidable rows: printed → re-verified	undecidable: printed → re-verified
---	---		---	---	---
release (the benchmark's own release)	**10/19 53 %**	≈	**14/19**	10/19 → **14/19**	—
protocol (the official protocol)	**13/19 68 %**	≈	**14/19**	13/19 → **14/19**	—
yolo_dist (the generic YOLO distribution package)	**4/19 21 %**	≈	**2/19**	4/15 ≈ 27 % → **2/17 ≈ 12 %**	4 → **2 rows**
reported (the layer actually reported)	**12-14 of the 19 ≈ 63- 74 %**		**10-11 of the 19**	12/15 = 80 % → **10/16- 11/17 ≈ 63- 65 %**	4 → **2-3 rows**

> **What a count here is, and what a "yes" means.** Every row is a **single rater's judgement**

> (§6.1), so the counts are printed as counts — the percentages beside them are a reading aid, not a

> rate estimated from independent coding. A "yes" under any

> unit means the reported number is **not a sample from an independent split**; it does **not** mean

> the benchmark is broken or the experiment dishonest, and it does not license a generalization claim.

These 19 rows were coded by one rater — 28 literal judgements (§10, defect 5). The five blind coders **independently reach 19 of 30 (63.3 %)** corrected cells, eleven unanimously, while **26 of 76** cells stay **undecidable**; a **non-blind, model-assisted** pass agrees at κ **0.874** and **moves two rows** (Supplementary S12). reported means **the number the literature prints**: it is printed as a range, **12–14 of the 19 $\approx$ 63–74 %**, with the **80 %** column the lower end over the **12/15** decidable rows, and n_a rows counted in the **19** but not the numerator (Supplementary S12).

6.2 "Same directory" has four levels, and the audit reported only the narrowest

The levels, the rows each admits and the resulting shift are in **Supplementary S12**, Table S12.

6.3 We do not reproduce the original audit's two numbers, and we say why one rests on the narrowest reading of "same directory", the other on a stale snapshot; both recomputations are in **Supplementary S12**, Table S13.

6.4 One judgment the evidence does not support (corrected)

One classification correction is recorded in Supplementary S12 (D-Fire moves to the clean group).

6.5 Across frameworks: is the convention specific to YOLO?

Among the pinned default configurations we checked, only the YOLO lineage does this by default. At pinned commits Detectron2 (TEST.EVAL_PERIOD = 0), MMDetection, DETR and PaddleDetection **report the final epoch instead** (file-by-file evidence, 25 configuration files: S12, Table S14; **a targeted sample at pinned commits, not a census**, so "the convention" means that family's, not the field's). A **cross-lineage realization test** — a Faster R-CNN (ResNet-50 FPN) detector, a family that **reports the final epoch by default** — on VisDrone under the same three-way split, at **six seeds**, is **positive in all six runs, with a 95 % interval excluding zero (68.2 % realized; +0.0809 pp mean prem_val, 95 % CI [26.9 %, 123.5 %])** (values, interval and per-run table: **Supplementary S34**): the mechanism is **not confined to the YOLO lineage in this instance**. A **second detector family** (RetinaNet, both corpora) keeps the **direction** but not the strength

(Supplementary S34e). A second corpus, same protocol, same direction: 6/6 and 12/12 runs. This is **one detector, one protocol, two corpora, one setting** — **not a generalisation beyond the YOLO lineage** (Supplementary S34c). Its **68.2 %** is a rate over **six runs** on this probe's own split and is **not comparable** with §8.6's realization rate: different protocol, different denominator.

7. Finding 4: what "significant" means, and how the endpoint convention changes it

7.1 With $n = 3$ a permutation test cannot reject in principle

A power note for the cells that remain at $n = 3$. Recovering each Tier-2 cell's sd from its own printed 95 % CI gives **0.20–0.67 pp**; at $n = 3$ the required effect is **0.65–2.19 pp** at $\alpha = 0.05$, two-sided — **at or above every Tier-1 effect here** (+0.27 to +0.53 pp) — and 0.2–0.7 pp at $n = 10$. The nine cells at $n = 3$ are therefore **not testable at this design**, and are reported descriptively only. A **prefix- n table ($n = 3 \dots 10$) and the derivation with the non-central- t factor ($3 \cdot 26 \cdot \text{sd}$) are in S12**; the permutation floor $2/2^n$ binds at $n = 3$, not the t test.

For a sign-flip permutation test on n pairs the smallest attainable two-sided p is $2/2^n$: at $n = 3$ that is **0.25**, so such a cell has **no resolution at this n** ; after §5.1's escalations no headline cell (Tables 1–2) is $n = 3$ (the derivation and each n 's floor: Supplementary S12).

7.2 The two clean-protocol cells clear significance by $n = 6$ –8

The weaker endpoint reaches $*t^*$ -significance at $n = 6$ on `shwd2sf` ($p = 0.0164$) and $n = 7$ on `smoke2sf` ($p = 0.0400$); the exact permutation — floored at $2/2^n$ (0.25 at $n = 3$) — follows one step behind on shwd2sf and none on smoke2sf ($n = 8$ on both `selected` readings; Table S42 gives every prefix n , both endpoints). The clean-protocol seed extension that reaches $n = 10$, and the per-cell numbers, are in Supplementary S12** and Table S15.

> **First-report attribution (a convention we hold to)**. Three things here belong to the companion

> paper and are cited, not claimed: the two Δ values (**+1.434 / +0.507 pp**), the three-way remediation

> protocol itself, and the pre-registered multi-target replication (§9). What this paper reports first is

> **the selection premium (§5), the 2×2 realization of that premium on a disjoint split (§8.6–8.7), and

> the formalisation of the realization rate (§3.1)**. The full statement — including how the companion is

> named to the editor while it carries no identifier — is in **Supplementary S12**.

7.3 A cross-archive endpoint conflict on the same run (investigated — and itself a disclosure)

One run is recorded with two different endpoints in two archives — which checkpoint is recorded was never specified: **Supplementary S12**, Table S16. Taking the final-epoch entry (67.12) gives $\Delta = -1.490$, $p = 0.0141$. The best-epoch entry (70.06) reads $\Delta = -0.902$, $p = 0.2931$, **but had been computed with the run's 'last' reading**; corrected 2026-10-02 it becomes $\Delta = -1.490$, $p = 0.0141$, as released. So under the 'best' convention of §5 this cell is **significant**, and nothing printed here moves.

8. Finding 5: how far the selection premium transfers to an independent split

8.1 The protocol is the companion's; the measurement under it is this paper's

The three-way remediation protocol is **defined and reported in the companion paper** — which first reports that **the effect reproduces in direction while its magnitude falls** — so this section **cites it** rather than presenting the replication. **In this paper's own terms it is three things**: a `val` carved from the pool so that training never sees it; the target corpus's own held-out `test`, with a filename-level check that its images do not intersect training; and an endpoint **defined in writing before any run**. Its **FROZEN-HASH** stamp is stated in §9; the registration `text` itself is the companion's and is **author-side**, so this paper carries the stamp and the readings (**Tables S24–S27**), not the document. What this paper measures

under that protocol is the quantity §3.1 formalises: **how much of the 'val'-selected premium survives on a split no selection rule can touch**, and that measurement is this paper's.

8.2 The protocol's results on the companion's corpora: transferred

The protocol's results on the companion's corpora are the companion's first report. What this paper adds under the protocol is the measurement of §8.6–§8.7; the transfer (2026-09-21) is recorded in the module-transfer note of that date.

8.3 A direct measurement of data-order replication (not seed variance, and not the selection claim itself)

A direct measurement of data-order replication — reproduced inside this setting: adding replicates that permute the training file order lowers the mean by 9 %–29 %, each figure taken **against that cell's own $n = 6$ mean** and not against the pooled headline effect, the **seed-budget discipline** measured rather than assumed ([52]'s setting, §8.4). **What these replicates are.** The per-run randomness is a **deterministic permutation of the training file order** (`--shuffle-seed`, the number in the run's name), **not** the framework's **seed** argument (Supplementary S12, section 9.3, with the two corpora whose replays are identical across `--seed` values). The quantity is therefore **data-order replication**, not seed variance: the 9 %–29 % bounds the **file-order** component of the reported mean, **not** the variance between independent training seeds. That best-of-K selection inflates a reported score is **not new** (§8.6 positions the closest concurrent work, a dose–response over 2,047 datasets), so **what is measured here is the magnitude under a protocol that selects cells for significance first and expands seeds afterwards** (design and recomputation check: Supplementary S12, Table S18). **The title is deliberately the weaker, sample-scoped claim:** with three initializations the interval is wide, and extending to five does not narrow it (the gain SD reaches the data-order component's order), so the components are **consistent with, not established by**, these initializations.

8.4 The three variance components: the excluded one is the smaller in this sample — a sample-scoped ordering, not an established one

The three variance components, and the excluded one is the smaller in this sample: the single-run data-order component is **0.515 pp**, against initialization and seed components of the same order — the **same ranking** as the shuffle-order result of [52] (Supplementary S12), so we **reproduce the ordering inside this setting** rather than discover it. What was held fixed, and why this is **sample-scoped** — at five initializations the gain SD moves **0.146 → 0.439 pp** and its interval **overlaps** the data-order magnitude, so the two are of the same order rather than one clearly smaller — is in Supplementary S12 and Table S19.

8.5 Why "same-pipeline pairing" is a necessity, not a habit

Pairing is not neutrality: the paired design covers the **budget and data** axes only, and **not** the optimiser's (why, and the counterexample: **Supplementary S12**).

8.6 The premium's realization rate: the claim-level interval includes zero ($[-17.3 \%, 53.2 \%, P = 0.118]$); the point estimates below are descriptive (arm-cell median 17.6 %, five arm-cells, $n = 10$; run-weighted 23.3 %)

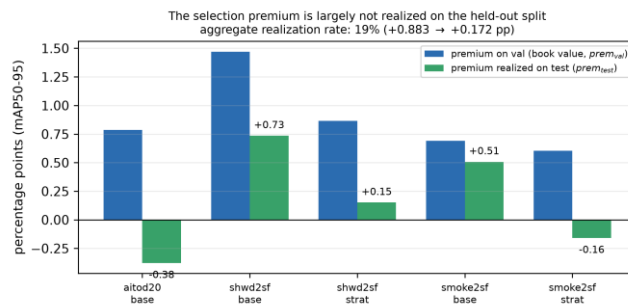


Fig. 2. The val-selected premium against what it realizes on the disjoint test split, per arm-cell; the title's aggregate is the plotted pool's — §8.6 names the pool Table 4 uses.

What this section claims, and what it does not. The rate below is the realization of the val-selected premium on a disjoint split **for the five arm-cells measured here, under the companion's protocol — not a constant**. The escalated cells reach a **pooled 55 %**, but the two batches are **not pooled** (different runs, different protocol, §8.1). "The convention realizes about a quarter of the val premium" would be false; "in these five cells, about a quarter survives" is what we measured.

How much of the gain selected on `val` survives on `test`?

Table 4. The premium selected on val versus what that same selection action realizes on the disjoint test split, per arm-cell (**an unbalanced pool** — two corpora contribute both arms, aitod20 only its baseline — so the arm-cell is the unit). Source: the **per-epoch `val` curves** (results.csv, for prem_val), xeval_20260916/matrix.csv, and this revision's two batches (x4_teval.csv on A, x4fill_20260925.csv on B, for prem_test). **The §10 memory-pressure bound (defect 10) does not arise here:** its hits are in §3.1's clean batch and §5.1's p_vistod15.

Cell	Arm	n	`prem_val`	**`prem_test`*	**realizatio

				*	n rate**
---	---	---	---	---	---
shwd2sf	baseline	10	+1.469	**+0.736**	**50 %**
shwd2sf	strategy	10	+0.865	**+0.152**	**18 %**
smoke2sf	baseline	10	+0.691	**+0.509**	**74 %**
smoke2sf	strategy	10	+0.605	**−0.157**	**−26 %**
aitod20	baseline	**10*	**+0.766*	**−0.218**	**−28 %**
		*	*		
aitod20	**strategy*	**10*	**+1.111*	**−0.543**	**−49 %**
†	*	*	*		

> **The inclusion rule, stated as a rule.** An arm-cell enters the pool iff its `test` side reached $n \geq 5$ (work/item2_final_v2_20260925.py), applied to every cell alike; **exactly five met it at the 2026-09-25 freeze** (50 runs). The aitod20 strategy row **qualifies under the same rule but completed after the freeze**, so it is printed here as the sixth row and kept out of the primary pool — **the `†` row, and the only one; both readings are reported and both intervals cover zero** (Table S20's note, §8.6b).

where $\text{prem_val} = V_{\text{max}} - V_{\text{final}}$ (selecting on val) and $\text{prem_test} = T_{\text{best}} - T_{\text{final}}$ (realized on test).

The three bases: run-level [8.4 %, 37.4 %], $P = 0.002$; five-cell [−17.3 %, 53.2 %], $P = 0.118$ (claim-level, **zero inside**); eleven-cluster [19.3 %, 60.4 %], $P < 0.001$ (a bootstrap tail proportion **at the 20,000-resample resolution**, i.e. below the floor this procedure can print).

Aggregate (runs weighted equally, $n = 50$): $\text{prem_val} +0.879$ pp $\rightarrow$ $\text{prem_test} +0.205$ pp. **At the claim level this is not significant:** the **claim-level interval** is the cell-level, five-cluster one, [−17.3 %, 53.2 %], $P(\text{rate} \leq 0) = 0.118$, zero inside. Descriptively, arm-cell median **17.6 %** and run-weighted **23.3 %** (span 2/5 negative: Table 4). On a difference scale the eleven clusters give **−1.367 pp**, 95 % CI [−1.59, −1.15] (five-cell denominators **+0.605 to +1.469 pp**). Re-aggregated: run-weighted **36.9 %**, arm-cell-equal **39.3 %**, 95 % CI [19.3 %, 60.4 %] — **completing the cell, not the weighting, is what moves it**: at $n = 10$ the three weightings agree, so **the fragile axis is pool composition**. **Leave-one-out 9.77 % to 34.16 %**. In two arm-cells the selection action **costs** rather than fails to pay off (smoke2sf strategy, aitod20 baseline); the escalated cells of §5.1 are **not pooled with these five** (different runs, different

protocol). The closest concurrent work [12], the winner's-curse check, the withdrawn $\sigma\sqrt{(2 \ln E)}$ form and each cell's corpus pair: **Supplementary S12**.

(the **best.pt**-versus-argmax residual, our withdrawn attribution and why the remainder is **unidentified**: **Supplementary S12**.)

The monotonicity is a correlation, not a causal claim, and at the arm-cell unit the nine-cluster reading is $\rho = -0.433$ ($p = 0.244$) — see **Fig. 3** and **Supplementary S12**: the $n = 80$ are the 81 runs with a measured **test** side less the one whose **prem_val** is 0. It has a named precedent: [53] gives validation overfitting (VO) its name and bounds the expected best observed loss by $E[\min L] \geq v^* - s_m\sqrt{(2 \log(Nq))}$, an optimism growing with the number of selection queries Nq .

Recomputation status. The readings were originally **not locally recomputable** (the archive lacks 6 of 7 corpora; §10, **defect 5**); a GPU 2×2 **evaluation** on **41** three-way runs of **YOLOv12n** [54] $\times$ {**best**, **last**} $\times$ {**val**, **test**} makes them so, **retraining not**, reproducing the archived **best** $\times$ **test** values to within **0.0047 pp** (median **0.0021 pp**) — the endpoint rule's positive control (**Supplementary S12**).

Three limitations: ① in the **41-run epoch sweep** only the two headline cells reach $n = 10$; **aitod20** contributes one run of 30 epochs and 6 checkpoints where others contribute **20** (hence 806, not 820), its Table 4 arm-cell a separate $n = 10$ batch; ② **closed: every retained checkpoint is now evaluated**, the *stability* reading is narrower than the grid; ③ κ is relative to each run's own final epoch, not causal. The 806-row evaluation is archived (xeval_perepoch_20260918/, with a SHA-256 list).

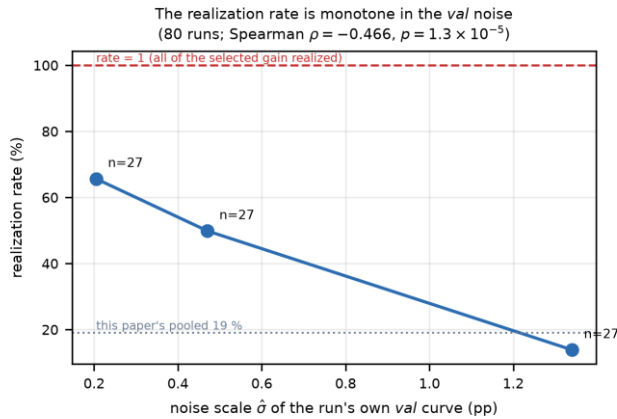


Fig. 3. Realization rate against the noise scale $\hat{\sigma}$ of the run's own `val` curve, over **80 runs** in three groups of 27 (Spearman $\rho = -0.466$, $p = 1.3 \times 10^{-5}$); dashed: rate = 1, dotted: §8.6's pooled reading.

Pool composition — and why the claim does not turn on it. Table 4 prints six measured arm-cells; adding the sixth (`aitod20 strategy`) moves the run-weighted rate **23.3 %** $\rightarrow$ **8.7 %** and the median **+17.6 %** $\rightarrow$ **-4.2 %**. **The inclusion rule is a rule, not a per-cell choice:** an arm-cell enters iff its `'test'` side reached $n \geq 5$, and exactly five met it at the freeze, where all three weightings coincide (**23.26 %**). The sixth row **qualifies but completed later**, so it is the sensitivity. **Every claim is interval-level and any scope gives the same verdict** — five-cell **[-17.3 %, 53.2 %]** ($P = 0.118$), six-cell **[-4.9 %, +21.3 %]**, eleven-cluster **[19.3 %, 60.4 %]** — the cell-level intervals **both cover zero**, so **no conclusion depends on the pool boundary**.

8.7 The per-epoch view: the realization rate is not a constant

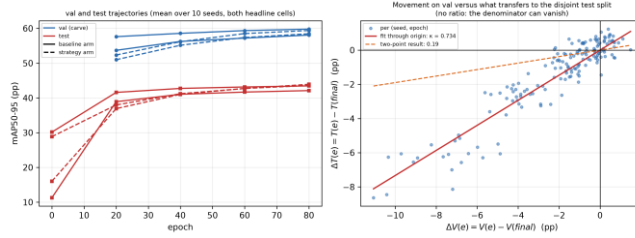


Fig. 4. **Left:** mean `val` and `test` trajectories over 10 seeds, both headline cells (baseline solid, strategy dashed). **Right:** per-(seed, epoch) `val` movement against what transfers to `test` — the ratio is not taken (its denominator can vanish); the fit through the origin is $\kappa = 0.734$.

One metric self-correction first. Our first version used $r(e) = \Delta T(e) / \Delta V(e)$, producing values such as **-860 %** and **+1031 %**: at late epochs $V(e) \approx V(\text{final})$, the **denominator vanishing** — **an ill-conditioned metric, not a finding**. We report only the non-dividing $\Delta V(e) = V(e) - V(\text{final})$ and $\Delta T(e) = T(e) - T(\text{final})$, summarized by the through-origin slope κ of ΔT on ΔV .

> **Tables S21–S23** $\rightarrow$ Supplementary §S8.

Slope κ (how much of the movement on `val` transfers to `test`). On the **complete curve** —

$n = 765$ (seed, epoch) points over the 41 runs — $\kappa = 0.708$, $R^2 0.937$, with a run-clustered (sandwich) 95 % CI [0.671, 0.745]; the within-run estimator gives 0.698. Two controls (Table S43): a per-run κ distribution, and a free-intercept fit whose intercept is indistinct from zero — so the through-origin constraint is supported, not assumed. Across the curve ΔV reaches

−8.1 pp and about 71 % transfers: early-training progress is real progress. §8.6's realized share measures the selection-relevant sliver near the peak, where ΔV is of order ± 0.2 pp — so

"most of the movement on `val` transfers" and "the little extra gained by selecting near the peak does not" are both true, and the premium is the residual of the transfer rate. And κ is not constant across arms either — in the direction the selection account predicts: baseline $\kappa = 0.600$ ([0.581, 0.619]) against strategy $\kappa = 0.767$ ([0.735, 0.800]), difference +0.168, 95 % CI [+0.130, +0.203].

A second result, and a self-correction the curve forces: the sign-of- Δ gap claim we wrote from four sampled epochs is too strong for one of the two cells (on the complete curve smoke2sf starts negative) and is narrowed here; per-checkpoint values: Supplementary S11.

The limitations of this section are in Supplementary S12.

9. The registered multi-target replication: reported in the companion paper

First report, criterion and verdict stay with the companion paper; nothing here is a claim of this paper. To read its registered table: ① it is a three-way split (train / selection / report); ② it ran under the protocol then in force, where the `val` side aliases the `test` half — the one fact §7 takes; ③ its criterion was frozen before any run existed (FROZEN-HASH 6a7eee7b3e34b15ce5adcba14cf7ea36) and its verdict entered is "criterion not met", deliberately not "hypothesis refuted" — the companion reads the outcome as "direction replicates, size does not". Record: Supplementary S24–S27 (source of truth: the companion's archive).

10. Our own defects (disclosed, not hidden)

#	Defect	Handling
---	---	---
1	The clean-protocol results document **cites a ledger file that does not exist in the sense claimed ** :	**We do not point at a replacement file ** (renaming would manufacture a new

	`_r10_final_stats.txt` nor the "real log" `_r10_stats_0403.txt` holds **any** headline value (6 probes each, **0/6 hits**) $\Rightarrow$ **no surviving stdout ledger**	false citation); the text says **"no stdout ledger"**, and the p-values are recomputable from <code>`pvalue_table_local.py`</code>. Detail: §S12
2	~the dota15 cell of <code>`selection_premium.py`</code> was not reproducible~ $\rightarrow$ **fixed** (stale root directory; the matcher assumed a seed suffix the run names do not carry); it now **reproduces M3's +1.12/+0.95/+0.93 value by value**	Stated in §5.3; the "3 seeds" note was **retracted on 2026-09-21** — the third run's seed is s42, the campaign's fixed shuffle seed. Detail: §S12
3	**`sio_b_results.csv` has no written endpoint convention**: <code>`t2_mask2mende_base100_s42n`</code> is recorded as final (67.12) in one archive and best (70.06) in another	Stated in §7.3, with the consequence of each reading (significant versus not)
4	The two headline scripts **originally depended on a rented pod** (<code>`import paramiko`</code> plus a plaintext password) — a paper about evaluation validity whose own analysis is not reproducible from the release	**Fixed**: both now **read the local archive**; the main text uses the local versions and lists the pod versions as cross-checks
5	**One script now recomputes the σ decomposition** — the data-order and initialization components from the archived per-run values, each asserted against the printed number — but it can only **cite** the augmentation component, whose record holds an SD and no per-run values; **no frozen effect-size rule file**; **no script generates <code>`benchmark_split_audit.md`</code> (hand-written, 28 judgments as literals, though its counts are recomputed)	The first is closed by script, with its one cited (not recomputed) component named; the other two stay labelled "manual / not archived"
6	The local archive **has no copy of 6 of the 7 corpora**, and of the three-way carve it holds **only the manifest** (<code>`split_defs_20260916/`</code>), not the other carve directories	**Partially repaired (2026-09-16, GPU run)**: the 41 runs' per-epoch curves, <code>`sio_b_results.csv`</code> and the **2×2 evaluation matrix** (164

		evaluations) are now local; **retraining remains non-recomputable**
7	The audit document has **internal inconsistencies** (nine items, found by our own census)	Each adjudicated: **3 misreadings by the inventory, 4 real defects, 1 convention risk, 1 classification correction (§6.4)**
8	Our own corpora contain `val == test`	**That is this paper's argument** , not an embarrassment to be hidden
9	**Our first per-epoch report claimed curve-wide stability from a four-point sample** ; the complete curve overturned it for one of the two cells (§8.7)	**Corrected in place, with the retracted sentence quoted** : the claim is **narrowed** , not deleted, and the grid's apparent uniformity is named as partly an artefact of where it sampled. Detail: §S12
10	**Memory pressure silently changes the training path** : under GPU memory pressure Ultralytics switches implementation **without changing the recorded configuration** — a batch reduction, or a fallback of label assignment to CPU that can fork the trajectory — and case (a) is **invisible in the standard artifacts** .	**Disclosed, not retracted; we do not call the affected batches “clean”.** Configuration-level hits: **2** places, **4** runs. **A reproducibility residual, not an event size** : same-config repeats are **bit-identical** (**0.000 pp** , four pairs, same machine and day), while the quantity a bound would be about — archived run against clean re-run — is **0.000 / 0.020 / 0.150 / 0.000 pp** (**Table S41**), so **≤ 0.05 pp** holds only inside the clean E = 50 batch and is **not extrapolated** ; without a no-event control the event effect is **unidentified** (such a control is itself **not

reproducible across
 dates**, **0.440 pp**),
 and elsewhere it sits
 inside the zero-event-
 control spread. The one
 clear case is $\Delta V_{\max}$
 +0.769 pp. Event
 counts, per-run
 consequences and the
 log fingerprints are in
 **Supplementary S12
 (defect 10)**.

11. Recommendations

Four checks before reporting, each with the artifact that discharges it. ①

Are `val` and `test` the same source? Make it a **hard pipeline check** on every dataset YAML (§4.2); the output is the resolved paths. ② If you must select on the reported split, say so and report a selection-free endpoint — **the 'final'-epoch endpoint, the only selection-free one this archive measures** — alongside — the step **enlarges** the arm difference (§5.1). ③ State the test convention: below $n = 6$ a permutation test has no resolution; print the $2/2^n$ floor beside every p . ④ Declare the counting unit behind every count, as Table 3's header does.

⑤ **A plateau is a post-hoc candidate, not a rule we validate.** Weight averaging is established [55]; what is *measured here* takes the **mean of the ranked top-5 'val' epochs' readings** (a reading average, not a weight average), capturing **70 %** of the realizable gain against the argmax's **30 %**, at the cost of **0.73 pp** below the peak — a post-hoc reading of our own 40 runs, untested as an intervention. The per-rule table and the candidate mechanism are in **Supplementary S11**, Measurement 3.

12. Limitations

- The audit reads **split definitions** (YAML, README, paper text), **no datasets downloaded**.
- For several benchmarks the "generic YOLO copy" is **per-project**, not canonical.
- Under the reported unit, **4 rows (AI-TOD, DIOR, Mendeley, D-Fire) are evidence-insufficient**; we do not guess. Counts are **row-equal over the 19**, and the grading enters the **error bar, not the count** (Table 3's second column).
- Two evidentiary differences (pod versus local) are listed side by side.

- The census is a **literature/release survey**, weaker than our **self-measurement** (§3).
- The **difficulty/selection decomposition** (§5.6) covers **2 of the 13 cells** (the headline two); the **identity** behind it is checked on **all 13** (work/gap_mechanism_20260916.txt) — what is scoped is the decomposition, not the identity.
- The intervention axis the headline comparisons use is **`lr0`** (**0.001 → 0.005**); two further orthogonal axes were run, one reproducing, as on **lr0**, the **corpus-dependent character** of the shift (**Supplementary S8, Table S40b**); the cross-lineage probe is **one detector, two corpora, and incomparable magnitudes**.
- **Four disclosed uncertainties** bound our claims: the 19-row audit is **single-rater**; the log sample is a **convenience sample**; the GPU memory-pressure effect is **bounded, not estimated** (**≤0.05 pp** where same-config pairs can be compared, fresh replicates being bit-identical; cross-date comparisons carry a **0.440 pp** drift); and the §8.3 replicates vary the **training file order**, not independent seeds. **And, not as an uncertainty: no interval here folds into §8.6's rate. Supplementary S12.**

13. Conclusions

A count is meaningless until its **unit** is named: the same nineteen benchmarks answer differently under each declared unit (Fig. 1, Table 3) — **10/19** under release, **13/19** under protocol, **4/19** under yolo_dist, and **12–14 of the 19** under reported (whose COCO-val reading is **13/19**). **Re-tallied after the 2026-10-09 first-party correction record, under the same per-column rule and with these printed counts frozen, the same four units read 14/19, 14/19, 2/19 and 10–11 of the 19** (Table 3, Supplementary S12) — the two **layer** units rise, the two **package and report** units fall, and the **majority** conclusion is unchanged (the weakest recomputed unit is still **10/19**). Three findings stand. **The most robust number is the transfer slope κ (§8.7): early `val` movement does transfer to `test`, and the premium is its residual.** The premium is positive in all four cells, the baseline gaining more, and the paired Δ gap **varies in sign** by corpus — so "the convention understates the effect" is not claimed. On five arm-cells at $n = 10$ it is realized on a disjoint split at **23.3 %**, interval containing zero. Two things do **not** hold: the $\sigma\sqrt{(2 \ln E)}$ scaling

was tested and **withdrawn**, and no power analysis was registered, so the nine $n = 3$ cells stay descriptive. What we ask of a detection paper is narrow: **name the counting unit, freeze in writing which checkpoint is reported, and state the interval at the claim's level**. First report is the companion's (§7.2).

Declarations

- **Data availability.** The **seven experiment corpora are open**; what the pack lacks is their **images**, so retraining is not recomputable (§10 defect 6). Separately, §5.5's **third-party logs are not redistributed** (hashes + pointers only), as **5 of their 6 sources state no licence**. The package — data, per-epoch matrix, verdicts, checkers, `MANIFEST_sha256.csv` — is at <https://github.com/YBP2005/P2.git> (accessed 2026-09-29). The reproduction package itself — data files, per-epoch matrix, verdict files, checkers, with their SHA-256 list in `MANIFEST_sha256.csv` — is deposited and openly available at <https://github.com/YBP2005/P2.git> (accessed 2026-09-29).

- **Recomputability boundary.** Every reported number is recomputable from the released per-run artifacts **without retraining**, each with its script; **three are not, and they are named**: the corpora and carve directories the archive **holds no copy of** (open corpora, images not in the pack; §10 defect 6), the conflicting-endpoint archives (§7.3), and the `aitod20` completion to $n = 10$.

- **Competing interests.** None declared.

- **One quantity, two spellings.** The $n = 10$ attainable floor is **exactly** $2/2^{10} = 0.001953$; the tables use the exact value, the abstract writes **0.002**. Same floor, spelling only.

- **Reproducibility.** Every number points to a local artifact (**artifact paths in this article and in the Supplementary are relative to the released `02\_release\_data/` directory**); **289 audit anchors** mechanically guard the claims, and six checkers (`audit_anchors_fupaper.py`, `check_crosscite.py`, `verify_pulled.py`, `gap_mechanism_20260916.py`, `verify_pr_forms_20260917.py`, `verify_submission_pack_20260917.py`) ship with the archive: **one runs unaided for a holder of the archive; four are author-side gates; and the remaining one -- the anchor audit itself -- is partly runnable, checking 270 of the 289 anchors from the released copy alone (the other 19 need the authors working-tree inputs)**. The archive is the one deposited above.

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> **Note on preprint citations.** Seventeen of the 55 entries give an arXiv edition; **three** also name a peer-reviewed venue and **fourteen** have none. Those fourteen cover four benchmark or dataset canonical releases, two methodological works and eight concurrent or tooling works; **all are cited for positioning only**, and every number comes from our own corpora or the third-party logs of §5.5. The full enumeration is in the supplement.

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